

Postulates, bra-ket notation and the qubit

Four rules, one new notation, and the simplest quantum system there is.

Dr. Abhijith Anandakrishnan · Assistant Professor
25AID432 · Odd Semester 2026-27

SESSION 10

The rules of the game

Everything in Unit 3 follows from four statements.

A postulate is a rule you accept, then test.

Nobody derives the postulates of quantum mechanics. Every experiment so far agrees with them.

You have used postulates before

School geometry. "Through two points there is exactly one straight line."

Nobody proves it. You assume it and build theorems on top.

Quantum mechanics works the same way.

The four postulates are assumed. What we check is every prediction built on them.

THE FOUR POSTULATES AT A GLANCE

Each one is Unit 2 mathematics with a physical meaning

	POSTULATE	IN ONE LINE	THE MATHS
1	State	a system is a unit vector	$\langle\psi \psi\rangle = 1$
2	Evolution	it changes by a unitary	$ \psi\rangle \rightarrow U \psi\rangle$
3	Measurement	outcome k has probability	$p(k) = \langle b_k \psi\rangle ^2$
4	Composite systems	joint space is a tensor product	$\mathbb{C}^2 \otimes \mathbb{C}^2$

Textbooks number these differently. The content is the same.

$\vec{v}$ from Unit 2 is $|v\rangle$ here.

Nothing new except the notation. It is due to Paul Dirac.

The ket is a column vector

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

Read it as “ket psi”.

The symbol inside is only a name. $|\psi\rangle$, $|v\rangle$, $|0\rangle$, $|\text{up}\rangle$ are all column vectors.

The bra is its dagger

$$\langle \psi | = |\psi\rangle^\dagger = (\alpha^* \quad \beta^*)$$

A row vector, with every entry conjugated.

Read it as “bra psi”.

Bra times ket is the inner product

$$\langle \phi | \psi \rangle = \langle \phi | \cdot | \psi \rangle = \phi_1^* \psi_1 + \phi_2^* \psi_2$$

Bra, then ket: a “bra-ket”, or bracket. The conjugate on the first argument is already built into the bra.

WORKED EXAMPLE 1

Two brackets, in both orders

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad |\phi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\langle\phi|\psi\rangle = \frac{1}{2} (1 \cdot 1 + 1 \cdot i) = \frac{1+i}{2} \quad \langle\psi|\phi\rangle = \frac{1}{2} (1 \cdot 1 + (-i) \cdot 1) = \frac{1-i}{2}$$

Swapping the order conjugates the answer: $\langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*$.

Forgetting to conjugate the bra

- **Transpose only**

$$\langle\psi| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \end{pmatrix}$$

Then $\langle\psi|\psi\rangle = \frac{1}{2}(1 + i^2) = 0$. A unit vector with length zero.

- **Transpose and conjugate**

$$\langle\psi| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \end{pmatrix}$$

Then $\langle\psi|\psi\rangle = \frac{1}{2}(1 + 1) = 1$, as it must be.

Ket times bra is a matrix

$$|0\rangle\langle 1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Column times row gives a grid. This is the **outer product** $|\psi\rangle\langle\phi|$.

$|0\rangle\langle 0|$ **keeps the part along** $|0\rangle$

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad |0\rangle\langle 0|\psi\rangle = |0\rangle \alpha = \alpha|0\rangle$$

$$|+\rangle\langle +| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad |+\rangle\langle +||0\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}|+\rangle$$

A **projector** $|b\rangle\langle b|$ does the projection of session 7 in one matrix.

The projectors of a basis add up to I

$$|0\rangle\langle 0| + |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$|+\rangle\langle +| + |-\rangle\langle -| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = I$$

Split a vector into its pieces along a basis, add them back, and nothing is lost.

TRANSLATION TABLE

Unit 2 → Unit 3

UNIT 2	UNIT 3	READ AS
$\vec{v}$	$ v\rangle$	ket
$\vec{v}^\dagger$	$\langle v $	bra
$\langle \vec{u}, \vec{v} \rangle$	$\langle u v\rangle$	bracket, inner product
$\ \vec{v}\ $	$\sqrt{\langle v v\rangle}$	norm
$\vec{u} \vec{v}^\dagger$	$ u\rangle\langle v $	outer product
$\vec{u}^\dagger A \vec{v}$	$\langle u A v\rangle$	"sandwich"

TRY THIS NOW · 5 MINUTES

Practise the notation

$$|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4i}{5}|1\rangle$$

1. Write $\langle\psi|$ as a row vector.
2. Compute $\langle\psi|\psi\rangle$.
3. Compute $\langle 0|\psi\rangle$ and $\langle 1|\psi\rangle$.
4. Write $|1\rangle\langle 0|$ as a matrix.

ANSWERS

Check yours

1. $\langle \psi | = \left(\frac{3}{5} \quad -\frac{4i}{5} \right)$
2. $\frac{9}{25} + \frac{16}{25} = 1$, a unit vector
3. $\langle 0 | \psi \rangle = \frac{3}{5}$ and $\langle 1 | \psi \rangle = \frac{4i}{5}$: the brackets pick out the entries
4. $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$

The bra is one method call

```
import numpy as np
psi = np.array([3/5, 4j/5])
bra = psi.conj()           # the bra, as a row
print(bra @ psi)           # <psi|psi> = 1
print(np.vdot(psi, psi))   # vdot conjugates the FIRST argument
print(np.outer([0, 1], [1, 0])) # |1><0|
```

What to do before next session

- Rewrite the answers of worked example 1 without the vectors, using only kets and bras.
- Learn the translation table.
- Recall how to normalise a vector (session 6).
- **Assignment Part B** is due at the end of this week.

SESSION 11

One qubit

The simplest quantum system: two levels, one unit vector.

A qubit is a unit vector in $\mathbb{C}^2$.

That is Postulate 1 for the smallest possible system.

Two complex amplitudes, one condition

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1$$

The complex numbers α and β are called **amplitudes**. Physically: an electron's spin, a photon's polarisation, two energy levels of an atom.

A superposition is a linear combination

$$\alpha|0\rangle + \beta|1\rangle$$

with both α and β nonzero.

The word sounds exotic. The mathematics is a sum of two vectors, exactly as in session 4.

What makes it quantum is what happens when you measure it (session 12).

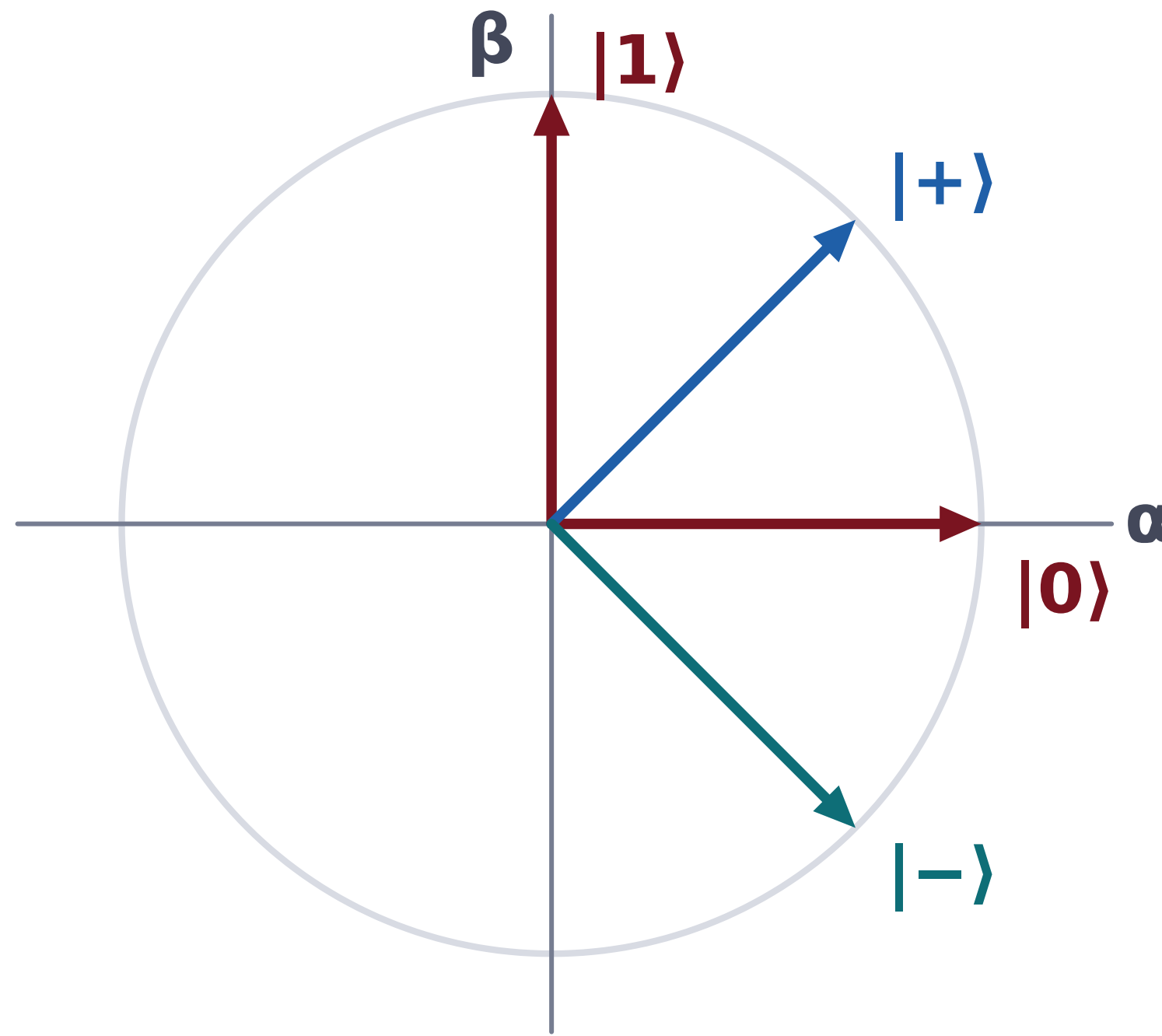
The standard states

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|\pm\rangle = \frac{1}{\sqrt{2}} (|0\rangle \pm |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$$

$$|\pm i\rangle = \frac{1}{\sqrt{2}} (|0\rangle \pm i|1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}$$

Four of them fit on a circle



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\alpha, \beta \text{ real, } \alpha^2 + \beta^2 = 1$$

Every such state is
a point on this circle.

$|0\rangle, |1\rangle$ are orthogonal; so are $|+\rangle, |-\rangle$:
each pair meets at 90° .

WORKED EXAMPLE 3

Normalise $3|0\rangle + 4i|1\rangle$

$$\|3|0\rangle + 4i|1\rangle\|^2 = |3|^2 + |4i|^2 = 9 + 16 = 25$$

$$|\psi\rangle = \frac{1}{5} (3|0\rangle + 4i|1\rangle) = \frac{3}{5}|0\rangle + \frac{4i}{5}|1\rangle$$

Divide by the norm, which is $\sqrt{25} = 5$.

WORKED EXAMPLE 4

Normalise $(1 + i)|0\rangle + |1\rangle$

$$|1 + i|^2 + |1|^2 = (1^2 + 1^2) + 1 = 3$$

$$|\psi\rangle = \frac{1}{\sqrt{3}} ((1 + i)|0\rangle + |1\rangle)$$

The squared modulus of $1 + i$ is 2. Squaring $1 + i$ itself gives $2i$, which is the wrong quantity.

Write $|0\rangle$ in the $\pm$ basis

$$\langle +|0\rangle = \frac{1}{\sqrt{2}}, \quad \langle -|0\rangle = \frac{1}{\sqrt{2}}, \quad \langle +|- \rangle = \frac{1}{2}(1 - 1) = 0$$

$$|0\rangle = \frac{1}{\sqrt{2}} (|+\rangle + |-\rangle), \quad |1\rangle = \frac{1}{\sqrt{2}} (|+\rangle - |-\rangle)$$

Components in a new basis are brackets, as in session 7.

$e^{i\gamma} |\psi\rangle$ and $|\psi\rangle$ are the same physical state.

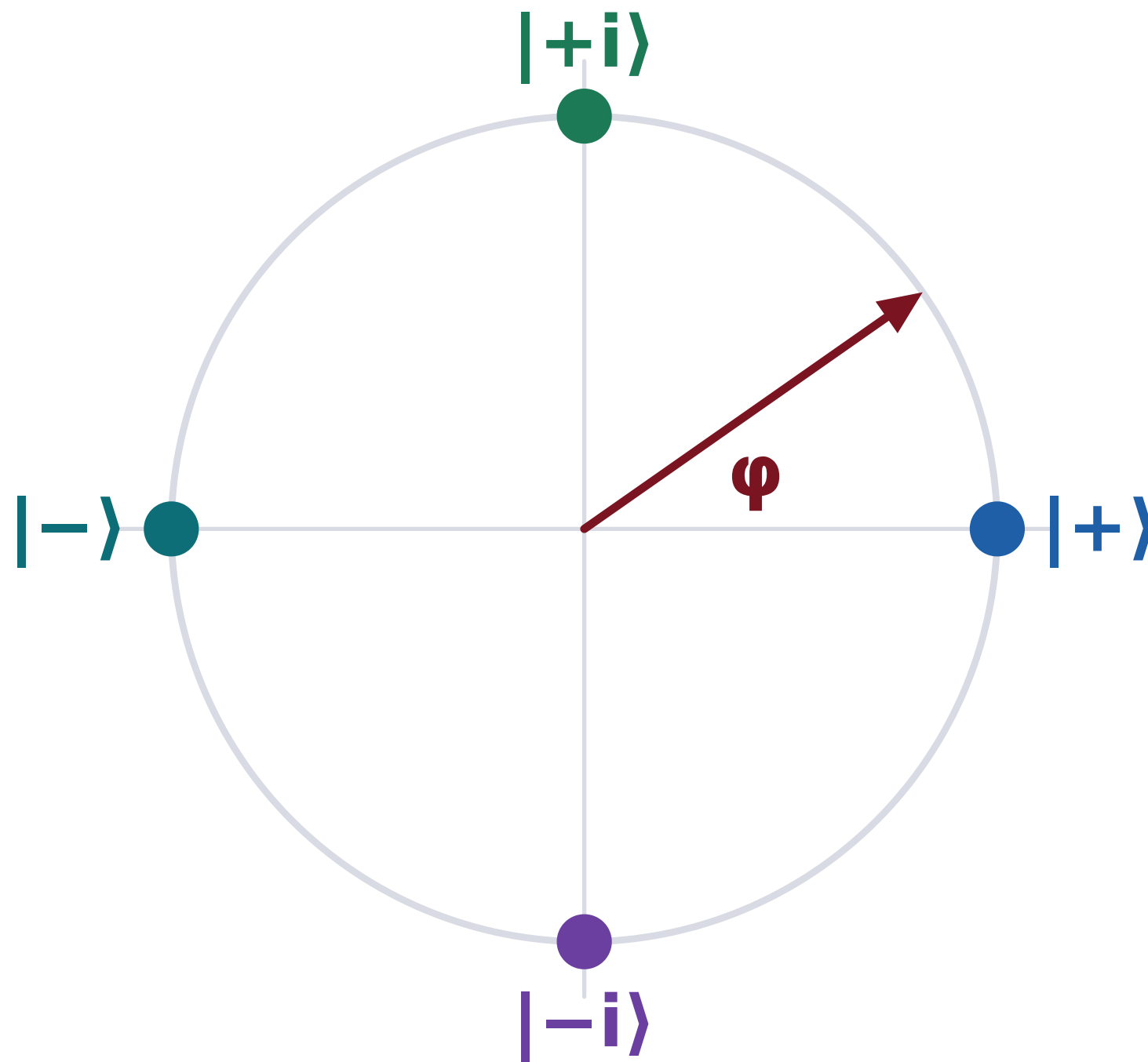
Multiplying the whole ket by a phase changes nothing you can ever measure. This is the **global phase**.

Every prediction contains $|e^{i\gamma}|^2 = 1$

$$|\langle b | e^{i\gamma} \psi \rangle|^2 = |e^{i\gamma}|^2 |\langle b | \psi \rangle|^2 = |\langle b | \psi \rangle|^2$$

Same probability for every outcome b , in every basis. Postulate 1 says the two kets are one state.

A phase between the amplitudes does matter



$$(|0\rangle + e^{i\varphi} |1\rangle) / \sqrt{2}$$

Same probabilities for 0 and 1
at every φ : $\frac{1}{2}$ and $\frac{1}{2}$.

Different φ , different state:
another basis tells them apart.

Global phase versus relative phase

- **Global phase**

Multiplies the **whole** ket.

So $-|0\rangle$ is $|0\rangle$, and $i|1\rangle$ is $|1\rangle$.

No measurement can tell them apart.

- **Relative phase**

Sits **between** the amplitudes.

Example: $|+\rangle$ and $|-\rangle$ differ by $e^{i\pi} = -1$ on $|1\rangle$ only.

A measurement in the $\pm$ basis tells them apart.

WORKED EXAMPLE 6

Same state or different?

$$\frac{1}{\sqrt{2}} (i|0\rangle + i|1\rangle) = i|+\rangle \quad \longrightarrow \quad \text{same state as } |+\rangle$$

$$\frac{1}{\sqrt{2}} (-|0\rangle + |1\rangle) = -|-\rangle \quad \longrightarrow \quad \text{same state as } |-\rangle$$

Method: pull out the phase of the first amplitude. Compare what is left.

MYTH

"A qubit is 0 and 1 at the same time"

- **The myth**

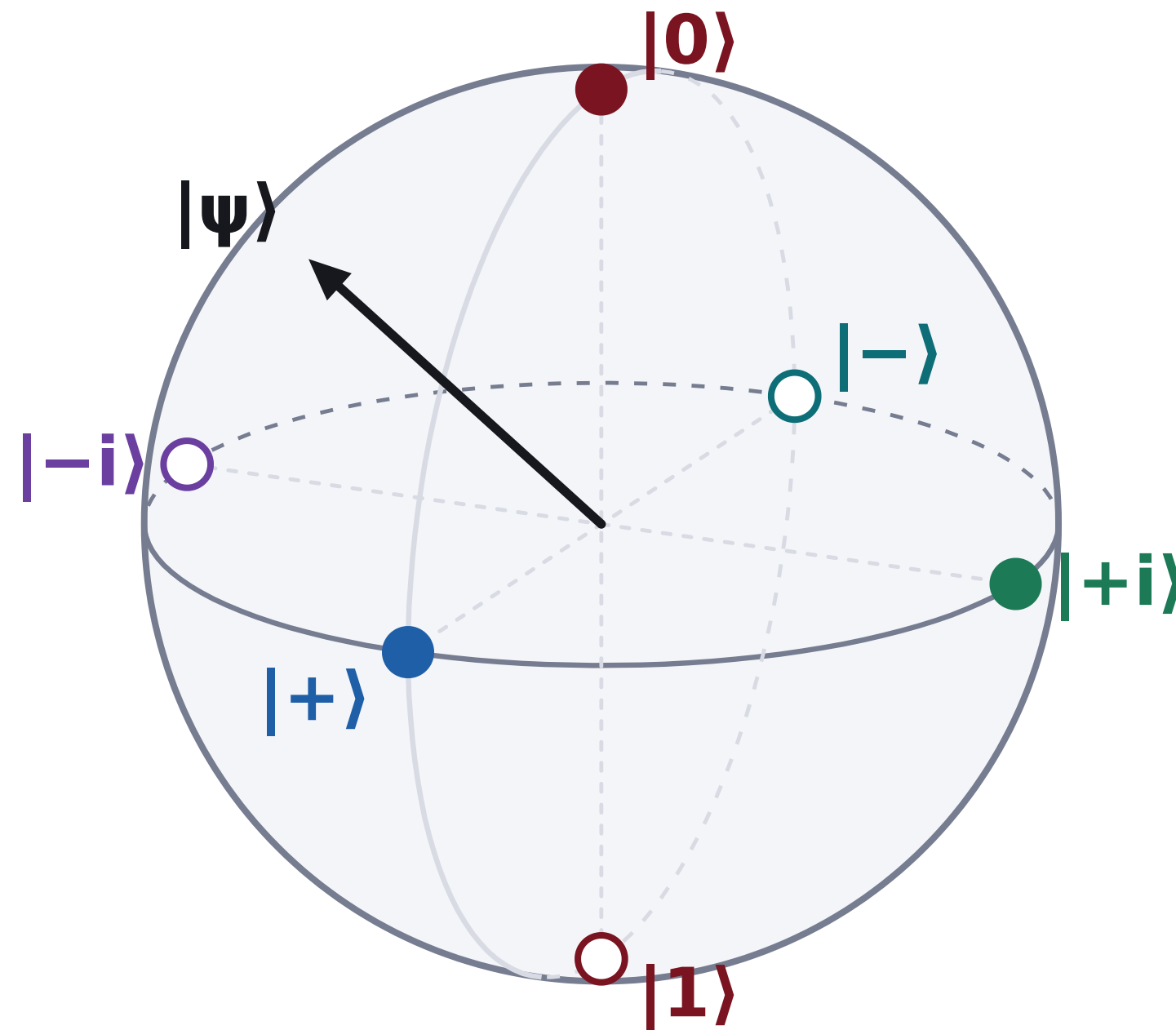
Reading $|+\rangle$ as "both 0 and 1", so that a qubit holds two values at once.

- **What the mathematics says**

The state $|+\rangle$ is one definite vector. Measure it in the $\pm$ basis and you get "+" every time.

Measure in the 0, 1 basis and you get **one** outcome, with probabilities.

The Bloch sphere: every qubit state is a point on a sphere



$$|\psi\rangle = \cos(\theta/2) |0\rangle + e^{i\varphi} \sin(\theta/2) |1\rangle$$

θ : angle down from $|0\rangle$
 φ : angle around the equator

Opposite points are
orthogonal states.

Hollow dot: behind the sphere

Why a sphere, and why half-angles

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

Remove the global phase and normalise: two real angles remain. The half-angle puts orthogonal states, such as $|0\rangle$ and $|1\rangle$, at opposite poles.

TRY THIS NOW · 5 MINUTES

Four quick decisions

1. Normalise $|0\rangle + i|1\rangle$. Which standard state is it?
2. Are $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $\frac{1}{\sqrt{2}}(-|0\rangle - |1\rangle)$ the same physical state?
3. Are $|+\rangle$ and $|-\rangle$ the same physical state?
4. Is $\frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle$ a valid qubit state?

ANSWERS

Check yours

1. $\frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) = | +i \rangle$
2. Yes. They differ by the global phase $-1 = e^{i\pi}$.
3. No. They differ by a relative phase on $|1\rangle$.
4. No: $\frac{1}{4} + \frac{1}{4} = \frac{1}{2} \neq 1$. Normalised, it becomes $|+\rangle$.

Normalise, and test for a global phase

```
import numpy as np
v = np.array([1, 1j])
psi = v / np.linalg.norm(v)           # normalise
print(psi)                            # [0.707, 0.707j]
a = np.array([1, 1]) / np.sqrt(2)
b = -a
print(abs(np.vdot(a, b)))              # 1.0: same state up to a phase
```

Two unit kets are the same state exactly when $|\langle a|b\rangle| = 1$.

What to do before next session

- Memorise the six standard states as vectors.
- Redo worked examples 3 to 6 without looking.
- Start Assignment Part C: build a NumPy array for each standard state.